Companion — openRCS user guide

This module controls an Analog Way **LiveCore** (Ascender, NeXtage, SmartMatriX Ultra) or **Midra** (Pulse2, Eikos2, Saphyr, SmartMatriX2, QuickMatriX, QuickVu) video processor from a Stream Deck or any other Bitfocus Companion surface, over the switcher's native TCP control protocol on port **10500**.

It's the Companion companion to the openrcs control surface: the same reverse-engineered protocol, driven from a surface instead of a browser.

The README covers installing the module. This guide is about building a page with it.

> **Before you rely on this:** the wire protocol has a smoke test with no dependencies that checks the encoding byte for byte on both platforms, and I've loaded the module against Companion's own base library: a bank-aware take sends the correct direction and encoding, a master recall sends the right pair of lines, and the feedbacks resolve. **It has not been tested from Companion against real hardware.**
>
> **If the connection comes up with no actions, no variables and no presets at all**, you're on a version that dies during startup. Upgrade to the newest release before you look at anything else.
>
> This module was built with AI assistance, directed and reviewed by a human author.

Configuring

- **Processor IP:** the switcher's address.
- **Platform:** LiveCore or Midra.

Pick the platform that matches the hardware. It decides more than a label: LiveCore ends commands with LF and Midra with CRLF, and the identity query is different too. Get it wrong and you don't get a helpful error, you get a switcher that ignores you.

Two things about that port:

> **The control protocol is unauthenticated.** Anyone who can reach port 10500 can drive the switcher. Keep it on a trusted network.

> **The device accepts very few concurrent control sessions.** If openrcs-server or the vendor's own client already has the connection, this module may not be able to connect at all. That's the switcher, not the module: close the other client.

Screens, groups and the 1-based count

"Screen/group" is **1-based**, and **an ungrouped screen is its own group**. So on a machine with no groups set up, screen 3 is group 3 and the numbering matches the front panel.

If you *have* set up groups, the numbering follows the groups, not the screens.

Take and Cut are bank-aware

It's worth getting your head round this before you build a transport row.

The module **keeps track of each group's on-air bank and takes to the other one**, rather than sending a fixed direction and hoping. That's why the take goes the right way even after somebody else has driven the desk from the vendor's UI, and it's also why the status query has to be working before a take means anything.

Step back puts a group back to its previous look. **T-bar** drives the transition bar by hand across its full 0–65535 range, so it's the action to put on a fader.

The actions

Take / Cut — screen/group	Transition or cut, with the direction resolved from the tracked bank.
T-bar	Manual transition, 0–65535.
Step back	Previous look.
Recall master / screen memory	Recall and take, or load to preview only. A load always names the preview bank, whatever another client left selected.
Load multiviewer layout	A stored layout memory onto a monitoring output (LiveCore). The monitor only.
Freeze input	Per input.
Output black	Per output.
Raw command	A protocol line directly, **mnemonic last** — e.g. `1,5000GCtba`. Supports Companion variables.

The thing to remember about raw commands is mnemonic last. The value comes first and is stuck straight onto the mnemonic with no separator. Get it the wrong way round and there's no error, just silence.

Feedbacks

- **On air (bank B / up):** the group's up bank is live.
- **Transitioning:** a take is in progress.
- **Available:** the group holds an active screen.

Put Available on every transport button. A take against a group with no active screen is a press that does nothing, and it looks the same as one that worked.

Presets

There are ready-made take and cut buttons for screens/groups 1–4 with on-air tally, recall buttons for master memories 1–8, and multiviewer layouts 1–8. Start from those rather than from blank buttons, because the tally's already wired up.

Troubleshooting

Symptom	Cause
No actions, variables or presets at all	The module died during startup. Upgrade to the newest release.
Cannot connect at all	Something else holds the switcher's control session — the vendor client, or openrcs-server. The device allows very few.
Commands are ignored, connection looks fine	Wrong **Platform**. The line terminator differs between LiveCore and Midra.
A take goes the wrong way	The tracked bank is stale. Check the connection has been up long enough to have read status.
A raw command does nothing	Mnemonic last, value first, glued: `1,5000GCtba`.
A transport button does nothing on one group	That group has no active screen. Add the *Available* feedback.

See also

- README: installing, and the full list of actions, feedbacks and variables
- `companion/HELP.md`: the same material, in Companion's help panel
- openrcs: the browser-based surface for the same protocol